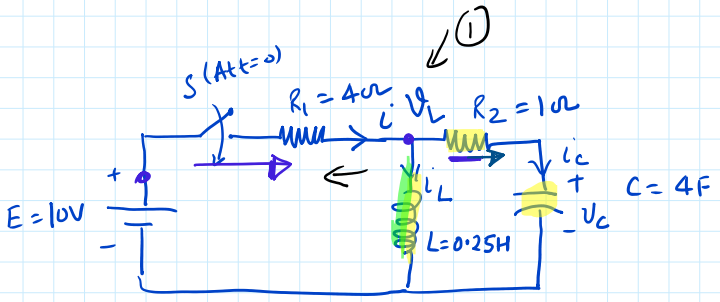
Example 4:

$v_C(t) = ?$
$i_L(t) = ?$
$i(t) = ?$

$i_L(0) = 0$
$v_C(0) = 0$



KCL at ①

$$\frac{v_L - E}{R_1} + i_L + i_C = 0$$

$$i_L + i_C = \frac{E - v_L}{R_1}$$

$$i_C = C\frac{dv_C}{dt} \quad (2)$$

$$v_L = i_C R_2 + v_C$$

$$v_L = R_2 C\frac{dv_C}{dt} + v_C \quad (3)$$

$$i_L = \frac{1}{L}\int_{-\infty}^{t} v_L(t)\,dt$$

$$i_L = \frac{1}{L}\int_{0}^{t} v_L(t)\,dt \quad (4)$$

$$i_L + i_C = \frac{E - v_L}{R_1}$$

$$\frac{1}{L}\int_{-\infty}^{t} v_L\,dt + C\frac{dv_C}{dt} = \frac{E - v_L}{R_1}$$

$$\frac{R_1}{L}\int_{0}^{t}\left(R_2 C\frac{dv_C}{dt} + v_C\right)dt + R_1 C\frac{dv_C}{dt} = E - \left(R_2 C\frac{dv_C}{dt} + v_C\right)$$

↓ Differentiate

$$\frac{R_1}{L}\left(R_2 C\frac{dv_C}{dt} + v_C\right) + R_1 C\frac{d^2 v_C}{dt^2} = 0 - \left(R_2 C\frac{d^2 v_C}{dt^2} + \frac{dv_C}{dt}\right)$$

$$(R_1 + R_2)C\frac{d^2 v_C}{dt^2} + \left(1 + \frac{R_1 R_2}{L/C}\right)\frac{dv_C}{dt} + \frac{R_1}{L}v_C = 0$$

$$\frac{d^2 v_C}{dt^2} + \frac{\left(1 + \frac{R_1 R_2}{L/C}\right)}{(R_1 + R_2)C}\frac{dv_C}{dt} + \frac{R_1/L}{(R_1 + R_2)C}v_C = 0$$

$$\frac{4/0.25}{(1+4)5} = 0.8$$

$$\frac{1 + \frac{4\times 1}{0.25/4}}{(1+4)4} = \frac{65}{20} = 3.25$$

$$\frac{d^2 v_C}{dt^2} + 3.25\frac{dv_C}{dt} + 0.8 v_C = 0$$

$$\frac{d^2v_c}{dt^2} + 3.25\frac{dv_c}{dt} + 0.8v_c = 0$$

$$\Downarrow$$

$$s^2 + 3.25s + 0.8 = 0 \Rightarrow \quad s_{1,2} = \frac{-3.25 \pm \sqrt{(3.25)^2 - 4\times 0.8}}{2}$$

$$= -0.2683, -2.9817$$

Thus

$$\boxed{v_c(t) = K_1 e^{-0.2683t} + K_2 e^{-2.9817t}}$$

Solve the constants K_1 and K_2 from the $v_c(0)$, $\frac{dv_c(0)}{dt}$ conditions.

At $t=0$, $v_c(0)=0 \Rightarrow$ $\quad K_1 + K_2 = 0 \Rightarrow K_1 = -K_2$

$$v_c(t) = K_1\left(e^{-0.2683t} - e^{-2.9817t}\right)$$

$$\left.\frac{dv_c(t)}{dt}\right|_{t=0} = K_1\left[(-0.2683) + 2.9817\right]$$

From KCL at node: $i_L + i_C = I = \frac{E - v_L}{R}$

$$\frac{1}{L}\int_{-\infty}^{t} v_L\,dt + C\frac{dv_c}{dt} = \frac{E - v_L}{R_1};$$

$$+ C\frac{dv_c}{dt} = \frac{E - v_L}{R_1}$$

Note

$v_L = v_c + R_2 i_c$

$= \left(v_c + R_2 C\frac{dv_c}{dt}\right)$

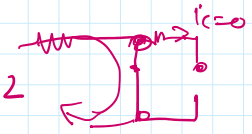
$$C\frac{dv_c}{dt} = \frac{E - \left(v_c + R_2 C\frac{dv_c}{dt}\right)}{R_1}$$

$$\left.(R_1 + R_2)C\frac{dv_c}{dt}\right|_{t=0} = E$$

$$\left.\frac{dv_c}{dt}\right|_{t=0} = \frac{E}{(R_1+R_2)C} = \frac{10}{(4+1)\times 4} = 0.5$$

$$\left.\frac{dv_c}{dt}\right|_{t=0} = K_1 \times 2.7134 = 0.5$$

$$K_1 = \frac{0.5}{2.7134} = 0.1842$$



at $t=0$ $\quad K_1 = \frac{0.5}{2.7134} = 0.1842$

$$v_c(t) = 0.1842\left(e^{-0.2683t} - e^{-2.9817t}\right)$$

$$i_C = C\frac{dv_c}{dt} = 0.7368\left(-0.2683e^{-0.2683t} + 2.9817e^{-2.9817t}\right)$$

$$i_L = \frac{1}{L}\int_{-\infty}^{t}\left(v_c + R_2C\frac{dv_c}{dt}\right)dt = \frac{1}{L}\int_{-\infty}^{0}\left(v_c + R_2C\frac{dv_c}{dt}\right)dt + \frac{1}{L}\int_{0}^{t}\left(v_c + R_2C\frac{dv_c}{dt}\right)dt$$

$$= 0 + \frac{1}{L}\int_{0}^{t}\left(v_c + R_2C\frac{dv_c}{dt}\right)dt$$

$$= \frac{1}{L}\int_{0}^{t} v_c\,dt + \frac{R_2C}{L}v_c$$

OR,

$$i_L + i_C = I = \frac{E - v_L}{R_1} = \frac{E - (v_c + R_2 i_C)}{R_1}$$

$$i_L = \frac{E - (v_c + R_2 i_C) - i_C}{R_1} = \frac{E}{R_1} - \frac{v_c}{R_1} - \left(1 + \frac{R_2}{R_1}\right)i_C$$

$$= \frac{10}{4} - \frac{0.1842}{4}\left(e^{-0.2683t} - e^{-2.9817t}\right) - 1.25\times 0.7368\left(-0.2683e^{-0.2683t} + 2.9817e^{-2.9817t}\right)$$

$$= 2.5 + 0.2e^{-0.2683t} - 0.5032e^{-2.9817t}$$

$$i_C(t) = 0.7368\left(-0.2683e^{-0.2683t} + 2.9817e^{-2.9817t}\right)$$

$$= -0.1977e^{-0.2683t} + 2.1969e^{-2.9817t}$$

$$i_L(t) = 2.5 + 0.2e^{-0.2683t} - 2.7e^{-2.9817t}$$

$$i = i_L + i_C = 2.5 + 0.0023e^{-0.2683t} - 0.5032e^{-2.9817t}$$

Thus note: $i_1(0) = 0$

Thus, note: $i_L(0) = 0$

$i_L(\infty) = 2.5\ A$

$v_c(0) = 0$

$v_L(\infty) = 0$

Example 5:

(a) Sketch and label the voltage v(t), and instantaneous power p(t) for an 8 μF capacitor whose current waveform in *milli-amps* is plotted in figure below. Assume that v(0) = 0.

(b) Sketch and label i(t) and instantaneous power p(t) for a 40 mH inductor, whose voltage waveform in *milli-volts* is plotted in Fig. 1. Assume that i(0) = 0.

Solution : Assignment 2, EE1100

Solution 1 (a)

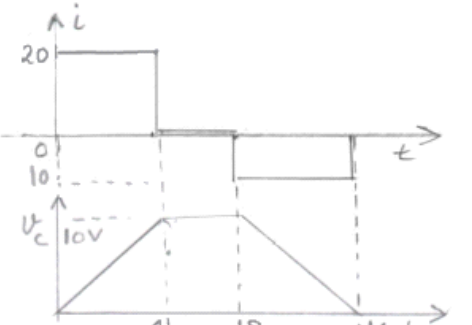
$$v_c(t) = \frac{1}{C}\int i_c\,dt = \frac{1}{C}\int_{-\infty}^{0} i_c\,dt + \frac{1}{C}\int_{0}^{t} i_c\,dt$$

$$= 0 + \frac{1}{C}\int_0^t i_c\,dt$$

$$= \frac{1}{8\times10^{-6}}\int_0^t (20\times10^{-3})\,dt \quad \text{for } 0 \le t \le 4 \text{ msec.}$$

$$= \frac{20}{8}t \quad \text{for } 0 < t \le 4 \text{ msec}$$

t is in msec.



For period $t \ge 4$ msec.

$$v_c(t) = v_c(t)\Big|_{t=4\text{ ms}} + \frac{1}{C}\int_4^8 0\cdot dt$$

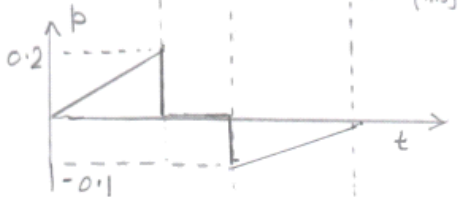
$$= 10\text{ V}$$

For period $4 \le t < 8$ msec

$$v_c(t) = v_c(t)\Big|_{t=8\text{ msec}} + \frac{1}{C}\int_8^{16} i_c\,dt$$

$$= 10 + \frac{1}{8\times10^{-6}}\cdot(-10\times10^{-3})\,t\Big|_{8\times10^{-3}}^{16\times10^{-3}}$$

$$= 10 - \frac{10}{8}\cdot(16-8) = 0\text{V}$$



Power $p(t)$ For $0 \le t < 4$ msec.

$$p(t) = v_c(t)\cdot i_c(t)$$

$$p(t) = 20\times10^{-3}\times\frac{20}{8}\cdot t$$

$p(t) = 0$ at $t = 0$

and $p(t) = 0.2\text{W}$ at $t = 4$ msec

For $4 \le t \le 8$ msec

$p(t) = 0$ as $i = 0$

For $8 \le t \le 16$ msec.

$$p(t) = v_c(t)\cdot i_c(t) = \left\{10 - \frac{10}{8}(t-8)\right\}\cdot(-10\times10^{-3})$$

$$= -10\times10^{-3}\left\{10 - \frac{10}{8}(t-8)\right\}$$

$\therefore p(t) = -0.1\text{ W}$ at $t = 8$ msec.

$p(t) = 0$ at $t = 16$ msec.

Soln. 1 (b).

$$i_L(t) = \frac{1}{L}\int v_L\,dt$$

$$= \frac{1}{40\times10^{-3}}\int_0^t 20\times10^{-3}\,dt\times10^{-3} \quad \text{with } t \text{ is in msec.}$$

$$= 0.5\times10^{-3}\,t \quad \text{for } 0 < t \le 4\text{ msec.}$$

$\therefore i_L(0) = 0\text{ A}$



Soln. 1(b).

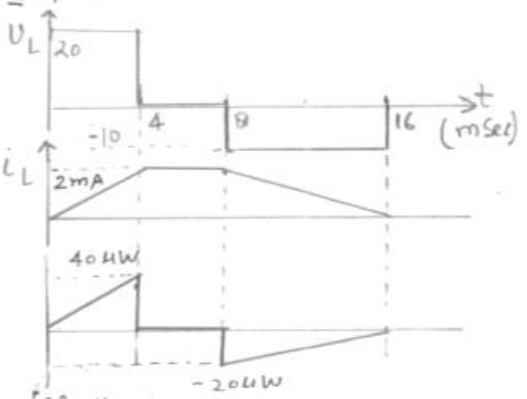
$$i_L(t) = \frac{1}{L}\int v_L\,dt$$

$$= \frac{1}{40\times10^{-3}}\int_0^t 20\times10^{-3}\,dt \times 10^{-3} \quad \text{with } t \text{ is in msec.}$$

$$= 0.5\times10^{-3}\,t \qquad \text{for } 0 < t \le 4 \text{ msec.}$$

$\therefore i_L(0) = 0$ A

$i_L(4\text{ msec}) = 2.0\times10^{-3}$ A



For $4 \le t < 8$ msec

$$i_L = i_L(4) + \frac{1}{L}\int v_L\,dt$$

$$= 2\times10^{-3} +$$

$$= 2\times10^{-3} \text{ A}$$

For $8 < t \le 16$ msec

$$i_L = i_L \text{ at } 8 \text{ msec} + \frac{1}{L}\int_8^t v_L\,dt$$

$$= 2\times10^{-3} + \frac{1}{40\times10^{-3}}\int_8^{16} -10\times10^{-3}\times10^{-3}\,dt$$

$$= 2\times10^{-3} - \frac{10^{-3}}{4}(t-8)\Big|_8^{16}$$

i_L at $t=8$ $= 2\times10^{-3}$ A

i_L at $t=16$ m s $= 2\times10^{-3} - 2\times10^{-2} = 0$

These are shown in figure.

Powers. $p(t) = v_L(t)\cdot i_L(t)$.

$$= 0.5\times10^{-3}\times 20\times10^{-3}\cdot t \qquad 0 < t \le 4 \text{ m s}$$

$$= 40\times10^{-6}\,t \text{ W}$$

$p(t) = 0$ ~~for~~ $4 \le t < 8$ m s

$$p(t) = \left\{2\times10^{-3} - \frac{10^{-3}}{4}(t-8)\right\}(-10\times10^{-3})$$

$= -20\times10^{-6}$ W at $t = 8$ m s

$= 0$ W at $t = 16$ m s